



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Main)

18-08-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	C	D	B	C	D	A	D	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	C	B	B	C	D	B	A	A	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.00	1250.00	220.00	45.00	0.80	30.00	46.00	440.00	5.00	1.80

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	C	C	B	C	A	C	C	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	D	B	C	D	C	D	B	D	C	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	250.00	100.00	9000.00	4.00	2.00	2.00	-1314.50	6080.00	2000.00	6.00

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	C	A	A	C	C	D	D	B	D
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	B	B	C	A	D	D	A	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	1.50	0.50	6.00	2.00	2.00	166.00	0.15 or 0.16	0.00	0.66 or 0.67	8.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (D)

Let acceleration of A be a_1 downwards and that of B is a_2 upwards and let T be the tension in the string. Then equations of motion of A and B gives

$$m_1 g - T = m_1 a_1 \quad \dots (i)$$

$$2T - m_2 g = m_2 a_2 \quad \dots (ii)$$

Equilibrium of pulley C gives

$$2T - T = 0$$

$$\Rightarrow T = 0$$

Substituting in (i) and (ii)

$$a_1 = g \downarrow \text{ and } a_2 = +g \downarrow$$

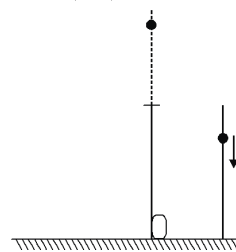
2. Ans (B)

At equilibrium $\frac{dU}{dr} = 0$

$$\frac{-12A}{r^{13}} + \frac{6B}{r^7} = 0$$

$$r = \left(\frac{2A}{B} \right)^{1/6}$$

3. Ans (C)

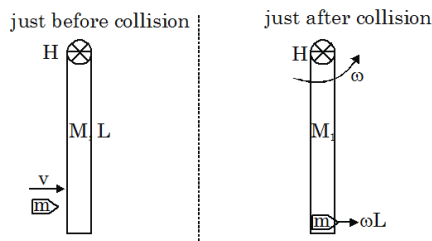


$$F_T = \lambda v^2 = \frac{m}{L} \left(\sqrt{2 \times g \frac{L}{2}} \right)^2$$

$$F_T = mg$$

$$F_{Net} = mg + \frac{mg}{2} = \frac{3mg}{2}$$

4. Ans (D)



$$(L_i)_H = (L_f)_H$$

$$mvL(\hat{v}) = m(\omega L)(L)\hat{v} + \frac{ML^2\omega}{3}\hat{v}$$

$$mv^2 = \left(\frac{3m+M}{3}\right)\omega v^2$$

$$\omega = \frac{3mv}{(3m+M)L}$$

5. Ans (B)

$$f_0 = \frac{1}{2\pi} \sqrt{\frac{mg\ell/2}{m\ell^2/3}}$$

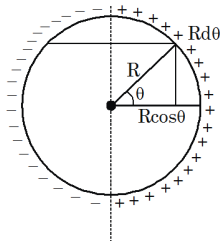
$$f_0 = \frac{1}{2\pi} \sqrt{\frac{3g}{2\ell}} \dots (i)$$

$$f' = \frac{1}{2\pi} \sqrt{\frac{(m/2)g(\ell/4)}{(\frac{m}{2})\frac{(\ell/2)^2}{3}}}$$

$$= \frac{1}{2\pi} \sqrt{\frac{1}{2} \times \frac{1}{4} \times 2 \times 4 \times 3 \cdot \frac{g}{\ell}}$$

$$= \frac{1}{2\pi} \sqrt{\frac{3g}{\ell}} = \sqrt{2} f_0$$

6. Ans (C)



$$dq = \lambda R d\theta$$

$$dq = \lambda_0 \cos\theta R d\theta$$

$$dp = dq \cdot 2R \cos\theta$$

$$= \frac{\lambda_0}{\pi/2} \cos\theta R d\theta \cdot 2R \cos\theta$$

$$p = \int dp = \int_{-\pi/2}^{\pi/2} 2R^2 \lambda_0 \cos^2\theta d\theta$$

$$= \frac{2R^2 \lambda_0}{2} \int_{-\pi/2}^{\pi/2} \cos 2\theta + 1 d\theta$$

$$HS-2/10 = R^2 \lambda_0 \left[\frac{\sin 2\theta}{2} + \theta \right]_{-\pi/2}^{\pi/2}$$

$$= R^2 \lambda_0 [\pi] = R^2 \lambda_0 \pi.$$

7. Ans (D)

$$(Te)_i = (Te)_\infty$$

$$-\frac{GMm}{R} + \frac{1}{2}m(3v_e)^2 = \frac{1}{2}m(v_\infty)^2$$

$$-\frac{GMm}{R} + \frac{1}{2}m9\left(\frac{2GM}{R}\right) = \frac{1}{2}m(v_\infty)^2$$

$$\frac{8GMm}{R} = \frac{1}{2}m(v_\infty)^2$$

$$\sqrt{\frac{16GM}{R}} = v_\infty = \sqrt{8}v_e = 2\sqrt{2}v_e$$

8. Ans (A)

Balance point = 40 cm

$$\therefore \frac{P}{40} = \frac{Q}{60}$$

$$P = \frac{2Q}{3} \dots (i)$$

when Q shunt with 10Ω resistance

$$P = Q' \left(Q' = \frac{10Q}{10+Q} \right)$$

$$\frac{2Q}{3} = \frac{10Q}{10+Q}$$

$$Q = 5\Omega$$

$$P = \frac{10}{3}\Omega$$

9. Ans (D)

$$U_i = \frac{1}{2}C_0 E^2 \dots (i)$$

On inserting a dielectric new capacitance is

$$C = kC_0 \dots (ii)$$

and new potential difference is $\{V = Ed\}$

$E' = f'_e$ where f_e is the electrostatic field. But

on inserting the dielectric, the new field f'_e

becomes $\frac{1}{k}$ times the original field.

$$\text{So } f'_e = \frac{f_e}{k}$$

$$\Rightarrow E' = \frac{E}{k} \dots (iii)$$

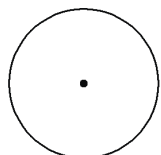
$$\text{So } U_f = \frac{1}{2}CE'^2$$

$$\Rightarrow U_f = \frac{1}{2}(kC_0)\left(\frac{E}{k}\right)^2$$

$$\Rightarrow U_f = \left(\frac{1}{2}C_0 E^2\right) \frac{1}{k}$$

$$\Rightarrow \Delta U = \frac{1}{2}C_0 E^2 \left(\frac{1}{k} - 1\right)$$

10. **Ans (A)**



$$B = Kt^2$$

$$\int \vec{E} \cdot d\vec{l} = \frac{d\phi}{dt} = \frac{AdB}{dt}$$

$$\Rightarrow E \cdot 2\pi r = \pi r^2 \cdot 2Kt$$

$$\Rightarrow E = Krt$$

Ring starts moving when torque due to electric field becomes greater than torque due to limiting friction

$$QE r = \mu Mgr$$

$$\Rightarrow Q \cdot Krt \cdot r = \mu Mgr$$

$$\Rightarrow t = \frac{\mu Mg}{QKr}$$

11. **Ans (A)**

Current sensitivity is given by $\phi/i = NAB/k$

$$(\phi/i)_1 = (30 \times 3.6 \times 10^{-3} \times 0.25)/k$$

$$(\phi/i)_2 = (42 \times 1.8 \times 10^{-3} \times 0.5)/k$$

$$\therefore (\phi/i)_1 / (\phi/i)_2 = 5/7 \text{ Ans.}$$

12. **Ans (C)**

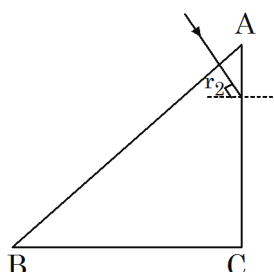
$$E = (10.2 \text{ eV}) (z - 1)^2$$

$$\frac{hc}{\lambda} = (10.2 \text{ eV}) (z - 1)^2$$

$$\frac{\lambda_2}{\lambda_1} = \frac{(z_1 - 1)^2}{(z_2 - 1)^2}$$

$$\frac{\lambda_2}{\alpha} = \frac{(56)^2}{(28)^2} \Rightarrow \lambda_2 = 4\alpha$$

13. **Ans (B)**



$$A = 90^\circ - \theta$$

$$r_2 = A = 90^\circ - \theta > \theta_c$$

$$\cos \theta > \sin \theta_c = \frac{6/5}{3/2} = \frac{4}{5} \text{ (}\theta_c \text{ is critical angle)}$$

$$\Rightarrow \theta < \cos^{-1} \frac{4}{5} = 37^\circ$$

14. **Ans (B)**

From snell's law

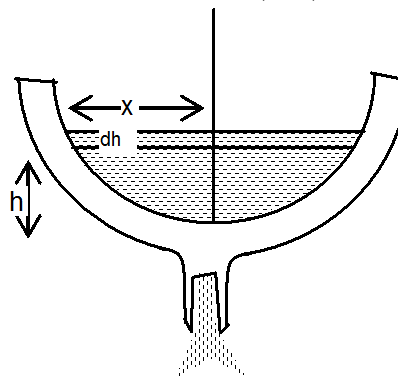
$$1 \sin 45^\circ = \sqrt{2} \sin \gamma$$

$$\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \sin \gamma$$

$$\gamma = 30^\circ$$

15. **Ans (C)**

$$\text{Hint } Q = av = \pi x^2 \left(\frac{dh}{dt} \right)$$



$$dt a \sqrt{2gh} = -\pi x^2 dh$$

$$x^2 = R^2 - (R - h)^2$$

16. **Ans (D)**

From Stefan-Boltzmann law

$$\text{Power} \propto T^4$$

From Wein's law

$$T \propto \frac{1}{\lambda_m}$$

$$\therefore \text{Power} \propto \frac{1}{\lambda_m^4}$$

$$P = \frac{K}{\lambda_0^4}$$

$$P_f = \frac{K}{\left(\frac{3\lambda_0}{4}\right)^4} = \left(\frac{4}{3}\right)^4 P$$

$$P_f = \frac{256}{81} P$$

17. **Ans (B)**

In this case, vernier constant

$$= \frac{1 \text{ mm}}{10} = 0.1 \text{ mm}$$

$$\text{Zero error} = 6 \times 0.1 = +0.6 \text{ mm}$$

$$\text{Correction} = -0.6 \text{ mm}$$

$$\text{Actual length} = (4.3 + 2 \times 0.01) + \text{correction}$$

$$= 4.32 - 0.06 = 4.26 \text{ cm}$$

18. Ans (A)

$$\begin{aligned} \text{Damping force } F &= -b \frac{dx}{dt} \\ &= -bx_0 [\omega_0 e^{-\omega_0 t} - \omega_0 (1 + \omega_0 t) e^{-\omega_0 t}] \\ &= bx_0 \omega_0^2 t e^{-\omega_0 t} \end{aligned}$$

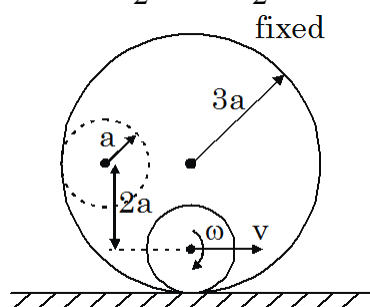
F is maximum when

$$\begin{aligned} \frac{dF}{dt} &= 0 \\ \Rightarrow bx_0 \omega_0^2 [e^{-\omega_0 t} - \omega_0 t e^{-\omega_0 t}] &= 0 \\ t &= \frac{1}{\omega_0} = \frac{1}{100} \text{ sec.} = 0.01 \text{ s} \end{aligned}$$

19. Ans (A)

Conservation of energy

$$Mg(2a) = \frac{1}{2} Mv^2 + \frac{1}{2} (Ma^2) (\omega)^2$$



For pure rolling $v = \omega a$

$$Mg(2a) = \frac{1}{2} Mv^2 + \frac{1}{2} (Ma^2) \frac{v^2}{a^2}$$

$$Mv^2 = Mg(2a)$$

$$v = \sqrt{2ga}$$

20. Ans (B)

$$V = \frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}$$

$$E = -\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k}$$

$$\frac{\partial V}{\partial x} = -\frac{1000}{x^2} - \frac{3000}{x^3} - \frac{1500}{x^4}$$

$$\frac{\partial V}{\partial y} = 0 \quad \& \quad \frac{\partial V}{\partial z} = 0$$

$$E(x, 0, 0) = \left(\frac{1000}{x^2} + \frac{3000}{x^3} + \frac{1500}{x^4} \right) \hat{i} \text{ V/m}$$

$$E(1, 0, 0) = (1000 + 3000 + 1500) \hat{i} \text{ V/m}$$

$$= 5500 \hat{i} \text{ V/m}$$

SECTION-II

1. Ans (1.00)

$$a_t = \alpha t$$

$$v = \frac{\alpha t^2}{2}$$

$$a_{\text{total}} = \sqrt{a_r^2 + a_t^2}$$

$$\tan 45^\circ = \frac{a_t}{a_r} = 1$$

$$\frac{\alpha t \times 4 \times 2}{\alpha^2 t^4} = 1$$

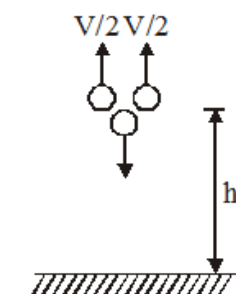
$$\text{at } t = 2 \text{ s}$$

$$\frac{\alpha(2) \times 4 \times 2}{\alpha^2 (2)^4} = 1$$

$$\alpha = 1 \text{ m/s}^3$$

2. Ans (1250.00)

$$h = V(10) + \frac{1}{2} g(10)^2$$



$$h = -\frac{V}{2}(20) + \frac{1}{2} g(20)^2$$

adding we get

$$h = 1250 \text{ m}$$

3. Ans (220.00)

Energy released = (B. E. of product – BE of reactant)

$$= (80 \times 7 + 120 \times 8 - 200 \times 6.5)$$

$$= 220 \text{ MeV.}$$

4. Ans (45.00)

In a closed tube, the first resonance takes place,

when the length of air column = $\lambda/4$

The wavelength of sound in air

$$\lambda = \frac{v}{n} = \frac{340}{340}$$

$$= 1\text{m} \text{ [As } v = 340 \text{ m/s and } n = 340]$$

Here length of first resonance can be given as

$$\frac{\lambda}{4} = 0.25\text{m} = 25 \text{ cm}$$

The length of second resonance

$$\frac{3\lambda}{4} = \frac{3}{4}\text{m} = 75 \text{ cm}$$

The length of third resonance

$$\frac{5\lambda}{4} = \frac{5}{4}\text{m} = 125 \text{ cm}$$

As the length of the tube is only 120 cm, hence third resonance is not possible.

Therefore, the minimum height of water required for resonance

$$= 120 - 75 = 45 \text{ cm}$$

5. Ans (0.80)

$$\text{The strain in the wire } \frac{\Delta \ell}{\ell} = \frac{2.0\text{mm}}{2.0\text{m}} = 10^{-3}$$

The stress in the wire = $Y \times \text{strain}$

$$= 2.0 \times 10^{11} \text{ Nm}^{-2} \times 10^{-3}$$

$$= 2.0 \times 10^8 \text{ Nm}^{-2}.$$

The volume of the wire

$$= (4 \times 10^{-6} \text{ m}^2) \times (2.0 \text{ m})$$

$$= 8.0 \times 10^{-6} \text{ m}^3.$$

The elastic potential energy stored

$$= \frac{1}{2} \times \text{stress} \times \text{strain} \times \text{volume}$$

$$= \frac{1}{2} \times 2.0 \times 10^8 \text{ N m}^{-2} \times 10^{-3} \times 8.0 \times 10^{-6} \text{ m}^3$$

$$= 0.8 \text{ J}.$$

6. Ans (30.00)

$$0.6 = \frac{\text{work done}}{Q_{\text{input}}} = \frac{Q_{\text{input}} - Q_{\text{reject}}}{Q_{\text{input}}} =$$

$$1 - \frac{Q_r}{Q_i}$$

$$0.6 = 1 - \frac{20}{Q_i}$$

$$Q_i = 50$$

$$W = Q_i - Q_r = 30 \text{ J}$$

7. Ans (46.00)

In this case we use

$$B_H = B_E \cos \theta$$

$$\Rightarrow B_E = \frac{B_H}{\cos \theta} = \frac{40 \times 10^{-6}}{\sqrt{3}/2}$$

$$\Rightarrow B_E = 46\mu\text{T}$$

8. Ans (440.00)

Position of minima

$$a \sin \theta = n\lambda$$

$$a \sin \theta = \lambda_1$$

$$\sin \theta_1 = \frac{\lambda_1}{a} \dots (i)$$

For maxima

$$a \sin \theta = \frac{3\lambda_2}{2} \text{ [approximately]} \dots (ii)$$

$$\sin \theta_1 = \frac{3}{2} \frac{\lambda_2}{a}$$

$$\lambda_1 = \frac{3}{2} \lambda_2$$

$$\lambda_2 = \frac{2}{3} \lambda_1 = 440 \text{ nm}$$

9. Ans (5.00)

$$\Delta \theta \geq 1.22 \frac{\lambda}{d}$$

$$R\Delta \theta = \frac{1.22 \times 550 \times 10^{-9}}{32 \times 10^{-3}} \times 24 \times 10^{-2} \text{ m}$$

$$= 503.25 \times 10^{-8}$$

$$= 5 \times 10^{-6} \text{ m} = 5\mu\text{m}.$$

10. Ans (1.80)

$$\frac{dT}{dt} = C (T - T_s)$$

$$3 = C(50 - 25)$$

$$C = \frac{3}{25}$$

When $T = 40^\circ\text{C}$

$$\frac{dT}{dt} = \frac{3}{25} (40 - 25) = \frac{45}{25} = 1.8^\circ \text{ C/min}.$$

PART-2 : CHEMISTRY

SECTION-I

1. **Ans (C)**

Spherical node = $\psi^2 = 0$

$$\sigma^2 - 4\sigma + 3 = 0 ; \sigma^2 - 3\sigma - \sigma + 3 = 0$$

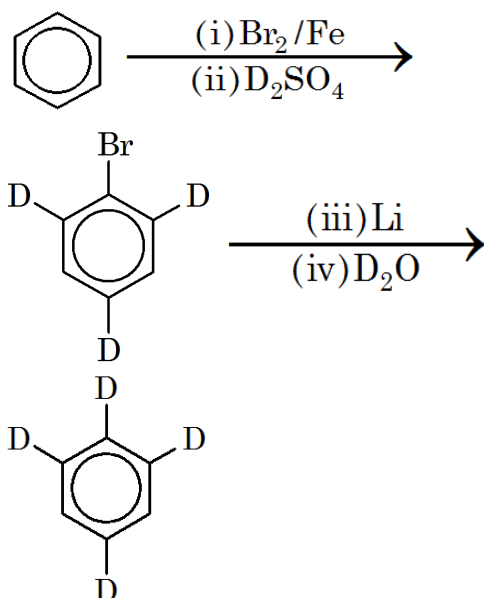
$$\sigma(\sigma - 3) - 1(\sigma - 3) = 0 ; (\sigma - 1)(\sigma - 3) = 0$$

$$\sigma = 1, 3 ; \sigma = \frac{2Zr}{a_0} = 1, 3 ; r = \frac{3}{2} \frac{a_0}{Z}, \frac{1}{2} \frac{a_0}{Z}$$

2. **Ans (A)**

Hit & Trial Method.

3. **Ans (C)**



4. **Ans (C)**

Aldol condensation.

6. **Ans (C)**

$\text{CH}_3\text{CH}_2\text{OH}$ shows H-bonding.

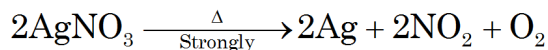
7. **Ans (A)**

Fact based.

8. **Ans (C)**

Fact based.

9. **Ans (C)**



(1) is possible on heating at low temperature.

(2) is not correct because Ag_2O is unstable.

(4) is not formed.

10. **Ans (A)**

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = 1.7 \times 10^{-10}$$

Precipitation will start when I.P.

exceeds K_{sp}

$$\text{I.P.} = [\text{Ag}^+][\text{Cl}^-] = 1.7 \times 10^{-10}$$

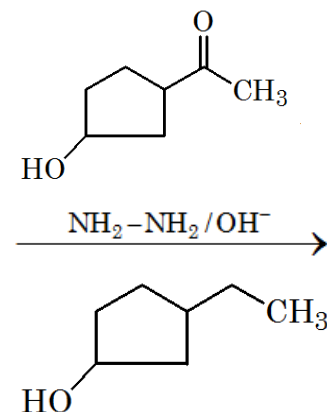
$$= \frac{1.7 \times 10^{-10}}{0.001} = \frac{1.7 \times 10^{-10}}{10^{-4}}$$

$$= 1.7 \times 10^{-6} \text{ mol L}^{-1}$$

11. **Ans (D)**

ccp has 8 tetrahedral voids within the unit cells.

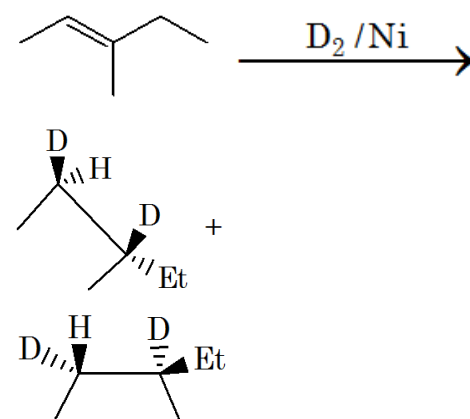
12. **Ans (B)**



13. **Ans (C)**

Fact based.

14. **Ans (D)**



15. **Ans (C)**

DNA is responsible for transmission of heredity characteristics.

16. **Ans (D)**

Glucose is used for silvering of mirrors.

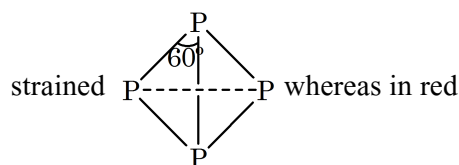
17. **Ans (B)**

(1) is not correct because it is a correct statement. Nitrogen forms multiple bonds because two N atoms can come sufficiently close together while P atom being large in size cannot come close together and hence does not form

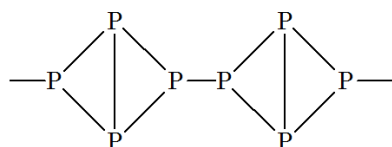
$P \equiv P$.

(2) is correct because it is an incorrect statement as the P-P single bond is more stronger than N-N because in N-N, there is more repulsive force between the non-bonding electron pair.

(3) is not correct it is a correct statement as in white phosphorus, the structure is highly



phosphorus, the structure is less strained



(4) is not correct because it is a correct statement as S-S single bond is a strong bond.

18. **Ans (D)**

(D) is correct because calcination is used to remove moisture, to remove organic matter and to convert carbonate ore to oxide.

19. **Ans (C)**

N is smaller in size, therefore most electronegative, Si is largest in size, therefore less electronegative than phosphorus. P is less electronegative than carbon due to bigger size.

20. **Ans (B)**

Photochemical smog is oxidising

SECTION-II

1. **Ans (250.00)**

$$\frac{P_C V_C}{RT_C} = \frac{3}{8}$$

$$\frac{73.89 \times V_C}{0.0821 \times 300} = \frac{3}{8}$$

$$V_C = \frac{1}{8} = 3b$$

$$b = \frac{1}{24}$$

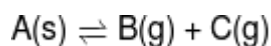
$$V_{\text{actual}} \times 4 = b$$

$$V_{\text{actual}} = \frac{b}{4} = \frac{1}{24 \times 4} \text{ for 1 mole}$$

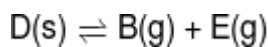
For 24 mole

$$V_{\text{actual}} = \frac{24}{24 \times 4} \times 1000 = 250 \text{ ml}$$

2. **Ans (100.00)**



$$K = 900 = P_1 (P_1 + P_2) \quad P_1 + P_2 = P_1$$



$$K = 1600 = P_2 (P_1 + P_2)$$

$$P_2 + P_1 P_2 \quad 2500 = (P_1 + P_2)^2$$

$$P_1 + P_2 = 50$$

Ans. 100 mm of Hg

3. **Ans (9000.00)**

$$q = 0$$

$$\Delta U = W = -P_{\text{ext}}(V_2 - V_1)$$

$$= -(40 - 50) = +40 \text{ bar L}$$

$$\Delta H = \Delta U + \Delta(PV)$$

$$= +40 + (P_2 V_2 - P_1 V_1) \text{ bar-L}$$

$$= +40 + (200 - 150) \text{ bar L}$$

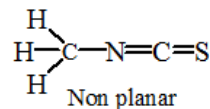
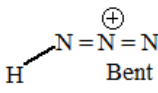
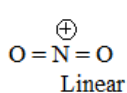
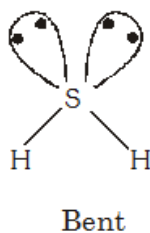
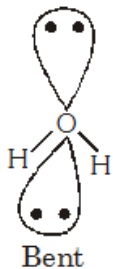
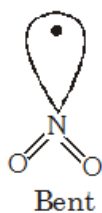
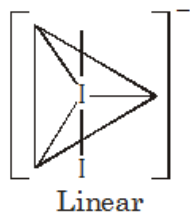
$$= 90 \text{ bar-L} = 9000 \text{ J}$$

4. **Ans (4.00)**

$$2^2 = 4.$$

5. Ans (2.00)

I_3^- and NO_2^+ are correct. The structure of these compounds are as



6. Ans (2.00)

$[Co(NH_3)_3(NO_2)_3]$ and $[Pt(NO_2)_3(NH_3)_3]^+$ can show Fac-mer isomerism since they are of type Ma_3b_3 .

7. Ans (-1314.50)

$$\Delta_r G^\circ = \Delta_f G^\circ (\text{Product}) - \Delta_f G^\circ (\text{reaction})$$

$$= -805 + 2(-228.6) - (52.3 + 0)$$

$$= -805 - 457.2 - 52.3 = -1314.5 \text{ kJmol}^{-1}$$

8. Ans (6080.00)

$$\pi = \frac{n_B RT}{V}; n_B = \frac{w_B}{M_B}$$

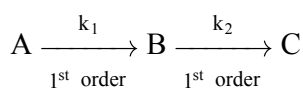
$$\pi = \frac{w_B}{M_B} \times \frac{RT}{V}$$

$$M_B = \frac{w_B}{V} \times \frac{RT}{\pi} = \frac{2 \times 0.08 \times 300 \times 760}{0.3 \times 20}$$

$$= 6080 \text{ gm mol}^{-1}$$

9. Ans (2000.00)

For a consecutive reaction



$$T_{[B]_{\max}} = \frac{\ln \frac{k_1}{k_2}}{k_1 - k_2}$$

10. Ans (6.00)

$$l = 2$$

$$Fe (26) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$$

$$Fe^{2+} (24) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$$

$$l = 2 \Rightarrow d$$

Electron present in ($l = 2$) d-orbitals = 6

PART-3 : MATHEMATICS

SECTION-I

1. Ans (C)

$$\sum_{r=1}^9 \sin^2 \frac{r\pi}{10} = \frac{1}{2} \sum_{r=1}^9 \left(1 - \cos \frac{2r\pi}{10} \right) = 5$$

2. Ans (C)

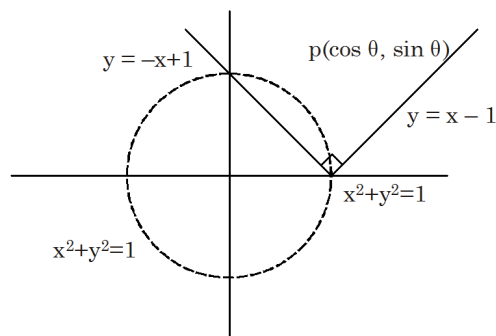
$$a_1 = 2 \cos \theta, b_1 = -2 \cot \theta$$

$$a_2 = 4 \sec \theta, b_2 = 4 \tan \theta$$

3. Ans (A)

$$P(\cos \theta, \sin \theta) \text{ lies on } x^2 + y^2 = 1$$

Clearly θ lies from $(0, \frac{\pi}{2})$



4. Ans (A)

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 5 & \lambda \end{vmatrix} \neq 0 \Rightarrow \lambda \neq 8$$

5. Ans (C)

Put $x = 1$ and -1 and then add.

6. Ans (C)

$$|A| = 1 \text{ and } A^{-1} = \frac{\text{adj}(A)}{|A|}$$

7. Ans (D)

$$f'(x) = 3x^2 + 2bx + a$$

Roots of $f'(x) = 0$ should be -3 and 2 .

8. **Ans (D)**

$$\text{limit} = e^{\lim_{x \rightarrow 0} \frac{1}{x} \log \left(\tan \left(\frac{\pi}{4} - x \right) \right)} = e^{-2}$$

9. **Ans (B)**

$$a^n + b^n = (a + b)^n$$

$$\therefore x^2 - 3 = 1 \Rightarrow x = \pm 2$$

10. **Ans (D)**

$$I = \int \frac{x^{-3}}{(4 + x^{-2})^6} dx \text{ put } 4 + x^{-2} = t$$

11. **Ans (A)**

$$f(x) = x^3 \quad x > 0$$

$$= 0 \quad x = 0$$

$$= -x^3 \quad x < 0$$

12. **Ans (B)**

$$I = \int_{-3}^3 x^8 \{x^{11}\} dx$$

$$I = \int_{-3}^3 x^8 \{-x^{11}\} dx$$

$$2I = \int_{-3}^3 x^8 dx = 2 \cdot 3^7$$

13. **Ans (B)**

$$\frac{dx}{dy} = \frac{x + 2y^3}{y} \Rightarrow \frac{dx}{dy} - \frac{1}{y}x = 2y^2$$

$$\text{I. F.} = e^{\int \frac{-1}{y} dy} = \frac{1}{y}$$

$$\therefore \frac{1}{y} \cdot x = \int \frac{1}{y} \cdot 2y^2 dy = y^2 + C$$

14. **Ans (C)**

$$\bar{X} = \frac{\sum_{r=0}^n r \cdot {}^n C_r}{\sum_{r=0}^n {}^n C_r} = \frac{n}{2}$$

$$\frac{1}{N} \sum f_i x_i^2 = \frac{1}{2^n} \sum_{r=0}^n r^2 {}^n C_r = \frac{n(n+1)}{4}$$

$$\text{Var}(x) = \frac{1}{N} \sum f_i x_i^2 - \bar{X}^2 = \frac{n}{4}$$

16. **Ans (D)**

$$\frac{1}{a} + \frac{1}{d} = \frac{1}{b} + \frac{1}{c}$$

Now by definition, R is Reflexive, Symmetric and Transitive.

17. **Ans (D)**

$$(p \rightarrow q) \equiv \sim p \vee q$$

$$\sim (\sim p \vee (q \wedge r)) \equiv p \wedge (\sim q \vee \sim r)$$

18. **Ans (A)**

Let P.V. of P, A, B and C be $\vec{p}$, $\vec{a}$, $\vec{b}$ and $\vec{c}$.

and O is circumcentre of equilateral ΔABC .

$$|\vec{b}| = |\vec{p}| = |\vec{a}| = |\vec{c}| = \frac{\ell}{\sqrt{3}}$$

$$\sum |\vec{PA}|^2 = 6 \cdot \frac{\ell^2}{3} - 2 \vec{p} \cdot (\vec{a} + \vec{b} + \vec{c}) = 2\ell^2$$

19. **Ans (B)**

Point of intersection of lines is (4, 3, 5)

Equation of plane $\perp$ to OP is

$$4x + 3y + 5z = 50$$

20. **Ans (A)**

$$\lim_{x \rightarrow 0} \frac{(1 - f(x))((f(x))^2 + f(x) + 1)}{5x^2}$$

$$= \frac{3}{5} \left(\frac{1 - \cos 2x \cdot \cos 4x \cdot \cos 6x \cdot \cos 8x}{x^2} \right)$$

Now use expansion of

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} \dots$$

SECTION-II

1. **Ans (1.50)**

$$\frac{\pi}{2} - \cos^{-1} x - \cos^{-1} x = \frac{\pi}{2} - \cos^{-1}(3x - 2)$$

$$2 \cos^{-1} x = \cos^{-1}(3x - 2)$$

$$\Rightarrow 2x^2 - 1 = 3x - 2 \Rightarrow 2x^2 - 3x + 1 = 0$$

$$\Rightarrow x = 1, \frac{1}{2}$$

2. **Ans (0.50)**

$$\text{Locus of mid point is } x^2 + y^2 - \frac{3}{5}x - \frac{4}{5}y = 0$$

3. **Ans (6.00)**

Two parabolas intersect at 4 points. Hence number of common chords are 4C_2 .

4. **Ans (2.00)**

$$x + \frac{1}{x} = 2 \cos \frac{\pi}{5}$$

$$\text{Let } x = e^{i \cdot \frac{\pi}{5}}$$

$$\Rightarrow x^{90} + \frac{1}{x^{90}} = 2 \cos 90 \frac{\pi}{5} = 2.$$

5. **Ans (2.00)**

$$\sum_{k=1}^{\infty} \frac{k}{2^k} \sum_{n=1}^{\infty} \frac{1}{2^n} = \sum_{k=1}^{\infty} \frac{k}{2^k}$$

6. **Ans (166.00)**

$$\text{Replace } x \rightarrow \frac{x+1}{3}$$

7. **Ans (0.15 or 0.16)**

$$\text{HHHH_} + \text{THHHH_} + \text{THHHH_} + \text{THHHH_}$$

$$\frac{1}{16} + \frac{1}{32} + \frac{1}{32} + \frac{1}{32} = \frac{5}{32}$$

8. **Ans (0.00)**

$$\begin{aligned}
 g(x) &= f(f(x) - 1) + f(5 - f(x)) \\
 &= f((x-1)^2 - 2) + f(6 - (x-1)^2) \\
 &= 2(x-1)^4 - 16(x-1)^2 + 32 \\
 &= 2((x-1)^2 - 4)^2 \geq 0, \forall x \in \mathbb{R}
 \end{aligned}$$

9. **Ans (0.66 or 0.67)**

$$\Delta = 2 \int_0^1 (\sqrt{x} - x^2) dx$$

10. **Ans (8.00)**

$$\ln f(x) = \ln(x-1) + \ln(x-2) + \dots + \ln(x-n)$$

$$f'(x) = f(x) \cdot \left[\frac{1}{x-1} + \frac{1}{x-2} + \dots + \frac{1}{x-n} \right]$$

$$f'(n) = (n-1)(n-2) \dots 3 \cdot 2 \cdot 1$$

$$5040 = (n-1)!$$

$$n = 8$$